Predicting House Prices

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House Price Prediction

Introduction

This project participates in the **Kaggle House Prices - Advanced Regression Techniques** competition. The goal is to predict the final sale price of residential homes in Ames, Iowa, using a comprehensive dataset containing 79 explanatory variables describing every aspect of the houses (from square footage and quality to features like fireplaces, garages, and pool quality).

Successful prediction of house prices is a classic regression problem with significant practical value in real estate analytics. This notebook demonstrates a complete end-to-end data science workflow — including extensive exploratory data analysis, feature engineering, handling of missing values, non-linear relationships, multicollinearity, and model diagnostics — culminating in a high-performing linear regression model.

Project Objectives

- Achieve a strong predictive performance on unseen test data (measured by Root Mean Squared Logarithmic Error - RMSLE)
- Build a well-specified and interpretable linear regression model
- Thoroughly validate model assumptions and document the modeling decisions
- Submit competitive predictions to the Kaggle leaderboard

The final model achieves an **Adjusted R^2 of 90.16%** on the training data and satisfies the core assumptions of linear regression to a high standard.

Its crucial to get an idea of the data set and a summary statistic is often useful to gain information on our data. We first read in the data and mutate our characted variables to factors for analysis purposes when we need them. For now we will not as the following analysis is going to only consider the numerical variables.

```
train <- read.csv("train.csv")

library(dplyr)
train <- train %>%
  mutate(across(where(is.character), as.factor))

train$MSSubClass <- as.factor(train$MSSubClass)
summary(train)
```

```
##        Id           MSSubClass     MSZoning     LotFrontage
##  Min.   :   1.0   20     :536   C (all):  10   Min.   : 21.00
##  1st Qu.: 365.8   60     :299   FV     :  65   1st Qu.: 59.00
##  Median : 730.5   50     :144   RH     :  16   Median : 69.00
##  Mean   : 730.5   120    : 87   RL     :1151   Mean   : 70.05
##  3rd Qu.:1095.2   30     : 69   RM     : 218   3rd Qu.: 80.00
##  Max.   :1460.0   160    : 63                  Max.   :313.00
##                   (Other):262                  NAs    :259
##     LotArea        Street      Alley      LotShape  LandContour  Utilities
##  Min.   :  1300   Grvl:   6   Grvl:  50   IR1:484   Bnk:  63    AllPub:1459
##  1st Qu.:  7554   Pave:1454   Pave:  41   IR2: 41   HLS:  50    NoSeWa:   1
##  Median :  9478               NAs :1369   IR3: 10   Low:  36
##  Mean   : 10517                           Reg:925   Lvl:1311
##  3rd Qu.: 11602
##  Max.   :215245
##
##    LotConfig    LandSlope   Neighborhood   Condition1     Condition2
##  Corner : 263   Gtl:1382   NAmes  :225   Norm   :1260   Norm   :1445
##  CulDSac:  94   Mod:  65   CollgCr:150   Feedr  :  81   Feedr  :   6
##  FR2    :  47   Sev:  13   OldTown:113   Artery :  48   Artery :   2
##  FR3    :   4              Edwards:100   RRAn   :  26   PosN   :   2
##  Inside :1052              Somerst: 86   PosN   :  19   RRNn   :   2
##                            Gilbert: 79   RRAe   :  11   PosA   :   1
##                            (Other):707   (Other):  15   (Other):   2
##    BldgType      HouseStyle   OverallQual      OverallCond      YearBuilt
##  1Fam  :1220   1Story :726   Min.   : 1.000   Min.   :1.000   Min.   :1872
##  2fmCon:  31   2Story :445   1st Qu.: 5.000   1st Qu.:5.000   1st Qu.:1954
##  Duplex:  52   1.5Fin :154   Median : 6.000   Median :5.000   Median :1973
##  Twnhs :  43   SLvl   : 65   Mean   : 6.099   Mean   :5.575   Mean   :1971
##  TwnhsE: 114   SFoyer : 37   3rd Qu.: 7.000   3rd Qu.:6.000   3rd Qu.:2000
##                1.5Unf : 14   Max.   :10.000   Max.   :9.000   Max.   :2010
##                (Other): 19
##   YearRemodAdd    RoofStyle       RoofMatl     Exterior1st   Exterior2nd
##  Min.   :1950   Flat   :  13   CompShg:1434   VinylSd:515   VinylSd:504
##  1st Qu.:1967   Gable  :1141   Tar&Grv:  11   HdBoard:222   MetalSd:214
##  Median :1994   Gambrel:  11   WdShngl:   6   MetalSd:220   HdBoard:207
##  Mean   :1985   Hip    : 286   WdShake:   5   Wd Sdng:206   Wd Sdng:197
##  3rd Qu.:2004   Mansard:   7   ClyTile:   1   Plywood:108   Plywood:142
##  Max.   :2010   Shed   :   2   Membran:   1   CemntBd: 61   CmentBd: 60
##                                (Other):   2   (Other):128   (Other):136
##    MasVnrType    MasVnrArea     ExterQual ExterCond  Foundation  BsmtQual
##  BrkCmn : 15   Min.   :   0.0   Ex: 52    Ex:   3   BrkTil:146   Ex :121
##  BrkFace:445   1st Qu.:   0.0   Fa: 14    Fa:  28   CBlock:634   Fa : 35
##  None   :864   Median :   0.0   Gd:488    Gd: 146   PConc :647   Gd :618
##  Stone  :128   Mean   : 103.7   TA:906    Po:   1   Slab  : 24   TA :649
##  NAs    :  8   3rd Qu.: 166.0             TA:1282   Stone :  6   NAs: 37
##                Max.   :1600.0                       Wood  :  3
##                NAs    :8
##  BsmtCond    BsmtExposure BsmtFinType1   BsmtFinSF1     BsmtFinType2
##  Fa :  45   Av :221      ALQ:220      Min.   :   0.0   ALQ:  19
##  Gd :  65   Gd :134      BLQ:148      1st Qu.:   0.0   BLQ:  33
##  Po :   2   Mn :114      GLQ:418      Median : 383.5   GLQ:  14
##  TA :1311   No :953      LwQ: 74      Mean   : 443.6   LwQ:  46
##  NAs:  37   NAs: 38      Rec:133      3rd Qu.: 712.2   Rec:  54
```

```
##                        Unf:430      Max.   :5644.0   Unf:1256
##                        NAs: 37                       NAs:  38
##    BsmtFinSF2        BsmtUnfSF       TotalBsmtSF      Heating     HeatingQC
##  Min.   :   0.00   Min.   :   0.0   Min.   :   0.0   Floor:   1   Ex:741
##  1st Qu.:   0.00   1st Qu.: 223.0   1st Qu.: 795.8   GasA :1428   Fa: 49
##  Median :   0.00   Median : 477.5   Median : 991.5   GasW :  18   Gd:241
##  Mean   :  46.55   Mean   : 567.2   Mean   :1057.4   Grav :   7   Po:  1
##  3rd Qu.:   0.00   3rd Qu.: 808.0   3rd Qu.:1298.2   OthW :   2   TA:428
##  Max.   :1474.00   Max.   :2336.0   Max.   :6110.0   Wall :   4
##
##  CentralAir Electrical     X1stFlrSF      X2ndFlrSF     LowQualFinSF
##  N:  95     FuseA:  94   Min.   : 334   Min.   :   0   Min.   :  0.000
##  Y:1365     FuseF:  27   1st Qu.: 882   1st Qu.:   0   1st Qu.:  0.000
##             FuseP:   3   Median :1087   Median :   0   Median :  0.000
##             Mix  :   1   Mean   :1163   Mean   : 347   Mean   :  5.845
##             SBrkr:1334   3rd Qu.:1391   3rd Qu.: 728   3rd Qu.:  0.000
##             NAs  :   1   Max.   :4692   Max.   :2065   Max.   :572.000
##
##    GrLivArea     BsmtFullBath     BsmtHalfBath       FullBath
##  Min.   : 334   Min.   :0.0000   Min.   :0.00000   Min.   :0.000
##  1st Qu.:1130   1st Qu.:0.0000   1st Qu.:0.00000   1st Qu.:1.000
##  Median :1464   Median :0.0000   Median :0.00000   Median :2.000
##  Mean   :1515   Mean   :0.4253   Mean   :0.05753   Mean   :1.565
##  3rd Qu.:1777   3rd Qu.:1.0000   3rd Qu.:0.00000   3rd Qu.:2.000
##  Max.   :5642   Max.   :3.0000   Max.   :2.00000   Max.   :3.000
##
##     HalfBath       BedroomAbvGr    KitchenAbvGr   KitchenQual  TotRmsAbvGrd
##  Min.   :0.0000   Min.   :0.000   Min.   :0.000   Ex:100      Min.   : 2.000
##  1st Qu.:0.0000   1st Qu.:2.000   1st Qu.:1.000   Fa: 39      1st Qu.: 5.000
##  Median :0.0000   Median :3.000   Median :1.000   Gd:586      Median : 6.000
##  Mean   :0.3829   Mean   :2.866   Mean   :1.047   TA:735      Mean   : 6.518
##  3rd Qu.:1.0000   3rd Qu.:3.000   3rd Qu.:1.000               3rd Qu.: 7.000
##  Max.   :2.0000   Max.   :8.000   Max.   :3.000               Max.   :14.000
##
##  Functional    Fireplaces    FireplaceQu   GarageType    GarageYrBlt
##  Maj1:  14   Min.   :0.000   Ex : 24     2Types :  6   Min.   :1900
##  Maj2:   5   1st Qu.:0.000   Fa : 33     Attchd :870   1st Qu.:1961
##  Min1:  31   Median :1.000   Gd :380     Basment: 19   Median :1980
##  Min2:  34   Mean   :0.613   Po : 20     BuiltIn: 88   Mean   :1979
##  Mod :  15   3rd Qu.:1.000   TA :313     CarPort:  9   3rd Qu.:2002
##  Sev :   1   Max.   :3.000   NAs:690     Detchd :387   Max.   :2010
##  Typ :1360                               NAs    : 81   NAs    :81
##  GarageFinish   GarageCars      GarageArea     GarageQual GarageCond PavedDrive
##  Fin:352      Min.   :0.000   Min.   :   0.0   Ex :   3   Ex :   2   N:  90
##  RFn:422      1st Qu.:1.000   1st Qu.: 334.5   Fa :  48   Fa :  35   P:  30
##  Unf:605      Median :2.000   Median : 480.0   Gd :  14   Gd :   9   Y:1340
##  NAs: 81      Mean   :1.767   Mean   : 473.0   Po :   3   Po :   7
##               3rd Qu.:2.000   3rd Qu.: 576.0   TA :1311   TA :1326
##               Max.   :4.000   Max.   :1418.0   NAs:  81   NAs:  81
##
##    WoodDeckSF      OpenPorchSF     EnclosedPorch      X3SsnPorch
##  Min.   :  0.00   Min.   :  0.00   Min.   :  0.00   Min.   :  0.00
##  1st Qu.:  0.00   1st Qu.:  0.00   1st Qu.:  0.00   1st Qu.:  0.00
##  Median :  0.00   Median : 25.00   Median :  0.00   Median :  0.00
```

```
##  Mean   : 94.24   Mean   : 46.66   Mean   : 21.95   Mean   :  3.41  
##  3rd Qu.:168.00   3rd Qu.: 68.00   3rd Qu.:  0.00   3rd Qu.:  0.00  
##  Max.   :857.00   Max.   :547.00   Max.   :552.00   Max.   :508.00  
##                                                                      
##   ScreenPorch        PoolArea        PoolQC       Fence      MiscFeature
##  Min.   :  0.00   Min.   :  0.000   Ex :   2   GdPrv:  59   Gar2:   2  
##  1st Qu.:  0.00   1st Qu.:  0.000   Fa :   2   GdWo :  54   Othr:   2  
##  Median :  0.00   Median :  0.000   Gd :   3   MnPrv: 157   Shed:  49  
##  Mean   : 15.06   Mean   :  2.759   NAs:1453   MnWw :  11   TenC:   1  
##  3rd Qu.:  0.00   3rd Qu.:  0.000              NAs  :1179   NAs :1406  
##  Max.   :480.00   Max.   :738.000                                      
##                                                                        
##     MiscVal             MoSold           YrSold        SaleType    
##  Min.   :    0.00   Min.   : 1.000   Min.   :2006   WD     :1267  
##  1st Qu.:    0.00   1st Qu.: 5.000   1st Qu.:2007   New    : 122  
##  Median :    0.00   Median : 6.000   Median :2008   COD    :  43  
##  Mean   :   43.49   Mean   : 6.322   Mean   :2008   ConLD  :   9  
##  3rd Qu.:    0.00   3rd Qu.: 8.000   3rd Qu.:2009   ConLI  :   5  
##  Max.   :15500.00   Max.   :12.000   Max.   :2010   ConLw  :   5  
##                                                     (Other):   9  
##  SaleCondition    SalePrice     
##  Abnorml: 101   Min.   : 34900  
##  AdjLand:   4   1st Qu.:129975  
##  Alloca :  12   Median :163000  
##  Family :  20   Mean   :180921  
##  Normal :1198   3rd Qu.:214000  
##  Partial: 125   Max.   :755000  
## 
```

Now that our data only includes numerical variables (for model simplicity purposes and curiosity of the power of this subset), we can set up our first model.

```r
model <- lm(SalePrice ~ ., data = num_vars_df)
summary(model)
```

```
## 
## Call:
## lm(formula = SalePrice ~ ., data = num_vars_df)
## 
## Residuals:
##     Min      1Q  Median      3Q     Max 
## -497980  -16701   -1933   13823  308740 
## 
## Coefficients: (2 not defined because of singularities)
##                 Estimate Std. Error t value Pr(>|t|)    
## (Intercept)    1.560e+05  1.431e+06   0.109 0.913252    
## Id            -1.291e+00  2.217e+00  -0.583 0.560288    
## LotFrontage    5.264e+01  2.860e+01   1.840 0.065920 .  
## LotArea        4.499e-01  1.019e-01   4.416 1.08e-05 ***
## OverallQual    1.647e+04  1.200e+03  13.725  < 2e-16 ***
## OverallCond    4.806e+03  1.037e+03   4.633 3.94e-06 ***
## YearBuilt      3.151e+02  6.174e+01   5.104 3.77e-07 ***
## YearRemodAdd   1.607e+02  6.700e+01   2.399 0.016587 *  
```

```
## MasVnrArea     2.909e+01  6.020e+00   4.832 1.50e-06 ***
## BsmtFinSF1     2.010e+01  4.733e+00   4.248 2.30e-05 ***
## BsmtFinSF2     8.953e+00  7.155e+00   1.251 0.211016
## BsmtUnfSF      1.084e+01  4.253e+00   2.547 0.010957 *
## TotalBsmtSF           NA         NA      NA       NA
## X1stFlrSF      4.931e+01  5.850e+00   8.429  < 2e-16 ***
## X2ndFlrSF      4.147e+01  4.920e+00   8.428  < 2e-16 ***
## LowQualFinSF   1.789e+01  1.995e+01   0.897 0.369936
## GrLivArea             NA         NA      NA       NA
## BsmtFullBath   7.942e+03  2.638e+03   3.011 0.002652 **
## BsmtHalfBath   3.683e+02  4.140e+03   0.089 0.929118
## FullBath       3.151e+03  2.852e+03   1.105 0.269363
## HalfBath      -1.386e+03  2.702e+03  -0.513 0.608147
## BedroomAbvGr  -9.036e+03  1.704e+03  -5.304 1.31e-07 ***
## KitchenAbvGr  -2.447e+04  4.936e+03  -4.957 8.01e-07 ***
## TotRmsAbvGrd   5.860e+03  1.250e+03   4.688 3.02e-06 ***
## Fireplaces     3.637e+03  1.796e+03   2.025 0.043074 *
## GarageCars     1.048e+04  2.895e+03   3.621 0.000304 ***
## GarageArea     2.595e+00  9.832e+00   0.264 0.791851
## WoodDeckSF     2.601e+01  8.109e+00   3.207 0.001369 **
## OpenPorchSF    1.029e+00  1.537e+01   0.067 0.946635
## EnclosedPorch  1.328e+01  1.710e+01   0.777 0.437504
## X3SsnPorch     2.286e+01  3.184e+01   0.718 0.472869
## ScreenPorch    5.470e+01  1.743e+01   3.139 0.001731 **
## PoolArea      -4.008e+01  2.407e+01  -1.666 0.096014 .
## MiscVal       -1.977e-01  1.885e+00  -0.105 0.916476
## MoSold         2.957e+01  3.496e+02   0.085 0.932610
## YrSold        -5.777e+02  7.115e+02  -0.812 0.417022
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 35250 on 1426 degrees of freedom
## Multiple R-squared:  0.8076, Adjusted R-squared:  0.8031
## F-statistic: 181.3 on 33 and 1426 DF,  p-value: < 2.2e-16
```

There are a lot of statistically insignificant variables which we will remove one by one or in sets if the p-value is very large, as they will never be significant, so anything aboove 0.5 will be removed right away and also log SalePrice as this is common practice given the exponential nature of house prices.

```r
model1 <- lm(SalePrice ~ ., data = num_vars_df[,-c(1,12,16)] )
#summary(model1)
model2 <- lm(SalePrice ~ ., data = num_vars_df[,-c(1,12,16,33,34,35,28,18)] )
#summary(model2)
model3 <- lm(SalePrice ~ ., data = num_vars_df[,-c(1,12,16,33,34,35,28,18,26,20)] )
#summary(model3)
model4 <- lm(SalePrice ~ ., data = num_vars_df[,-c(1,12,16,33,34,35,28,18,26,20,30,29,15)] )
#summary(model4)
model5 <- lm(log(SalePrice) ~ ., data = num_vars_df[,-c(1,12,16,33,34,35,28,18,26,20,30,29,15)] )
#summary(model5)
model6 <- lm(log(SalePrice) ~ ., data = num_vars_df[,-c(1,12,16,33,34,35,28,18,26,20,30,29,15,2,8,21)] )
summary(model6)
```

```
##
```

```
## Call:
## lm(formula = log(SalePrice) ~ ., data = num_vars_df[, -c(1, 12,
##     16, 33, 34, 35, 28, 18, 26, 20, 30, 29, 15, 2, 8, 21)])
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.01055 -0.06767  0.00578  0.07979  0.51970
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3.048e+00  5.259e-01   5.796 8.35e-09 ***
## LotArea       2.035e-06  4.274e-07   4.761 2.12e-06 ***
## OverallQual   8.218e-02  4.910e-03  16.737  < 2e-16 ***
## OverallCond   4.913e-02  4.275e-03  11.492  < 2e-16 ***
## YearBuilt     2.746e-03  2.262e-04  12.139  < 2e-16 ***
## YearRemodAdd  1.069e-03  2.750e-04   3.887 0.000106 ***
## BsmtFinSF1    8.774e-05  1.947e-05   4.507 7.09e-06 ***
## BsmtFinSF2    7.940e-05  2.984e-05   2.661 0.007889 **
## BsmtUnfSF     6.262e-05  1.772e-05   3.534 0.000422 ***
## X1stFlrSF     2.054e-04  2.415e-05   8.506  < 2e-16 ***
## X2ndFlrSF     1.660e-04  1.754e-05   9.465  < 2e-16 ***
## BsmtFullBath  5.429e-02  1.054e-02   5.149 2.98e-07 ***
## FullBath      2.731e-02  1.093e-02   2.498 0.012617 *
## KitchenAbvGr -1.028e-01  2.052e-02  -5.008 6.18e-07 ***
## TotRmsAbvGrd  2.028e-02  4.543e-03   4.463 8.70e-06 ***
## Fireplaces    4.503e-02  7.425e-03   6.065 1.69e-09 ***
## GarageCars    7.724e-02  7.179e-03  10.758  < 2e-16 ***
## WoodDeckSF    1.107e-04  3.374e-05   3.280 0.001061 **
## ScreenPorch   3.371e-04  7.246e-05   4.653 3.58e-06 ***
## PoolArea     -3.674e-04  9.977e-05  -3.682 0.000240 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1487 on 1440 degrees of freedom
## Multiple R-squared:  0.8632, Adjusted R-squared:  0.8614
## F-statistic: 478.2 on 19 and 1440 DF,  p-value: < 2.2e-16
```

We now have a model with statistically significant variables that explains 86.32% of the variation in logged house sale prices (Adjusted $R^2 = 0.8632$). This represents a strong linear model. However, before proceeding with interpretation or submission, we must verify that the model satisfies the key assumptions of linear regression. Only then can we be confident that the results are reliable and statistically valid.

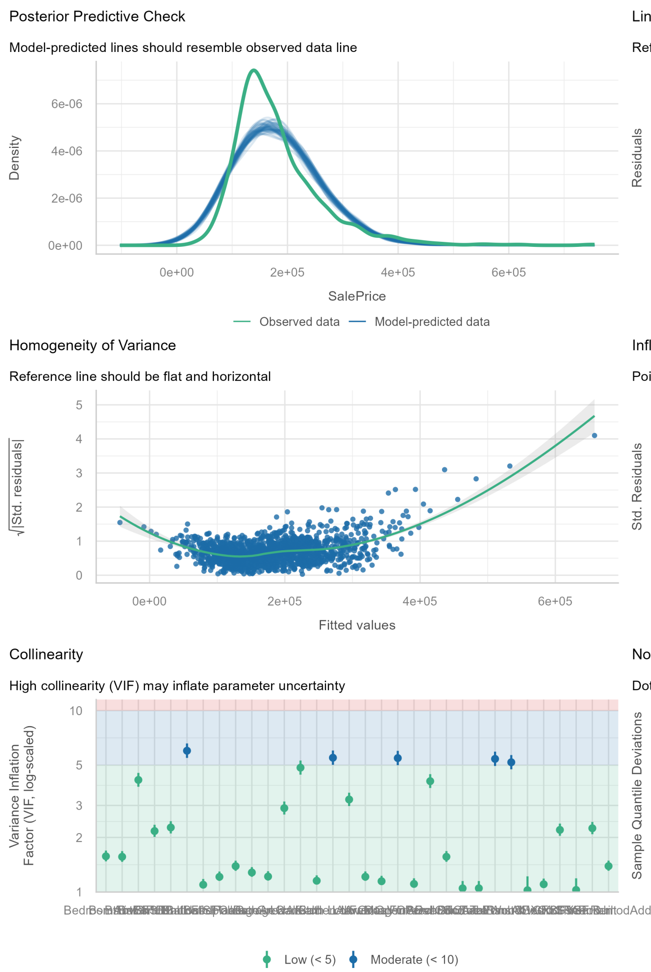
We must verify that the model satisfies the key assumptions of linear regression:

- **Linearity**: The relationship between the predictors and the response variable is linear.
- **Independence of errors**: The residuals are independent of each other.
- **Homoscedasticity**: The variance of the residuals is constant across all levels of the fitted values.
- **Normality of residuals**: The residuals are approximately normally distributed.
- **No severe multicollinearity**: The predictors are not highly correlated with each other.

In the following section, we perform a series of diagnostic checks to assess whether these assumptions hold.

```r
library(tidyverse)
library(performance)    # for nice checks
library(car)            # for VIF and other diagnostics
library(ggfortify)      # optional: ggplot2 versions of base plots
library(patchwork)      # for combining plots nicely
```

```
## Ignoring unknown labels:
## * size : ""
```



In the Posterior Predictive Checks section of the above plot, we can see that the model fits well for most regions of SalesPrice, except for around the median value of Sales price. This pairs well with the Linearity plot, as it is clear that the reference line is not linear and the points also deviate from the line, hence we should check for some non-linear relationships between our predictors and Sales price.

There is also evidence to suggest that there is non-equal variance (non-homogeneity of variance) as the reference line is clearly not flat, showing a quadratic shape, although the spread above and below the line is quite uniform, which is good.

There appears to an influential observation as shown in the Cook's Distance Contour plot, observation 299 should be removed as its residual standard deviation is too far from the expected deviation for us to keep it without it influencing our model negatively.

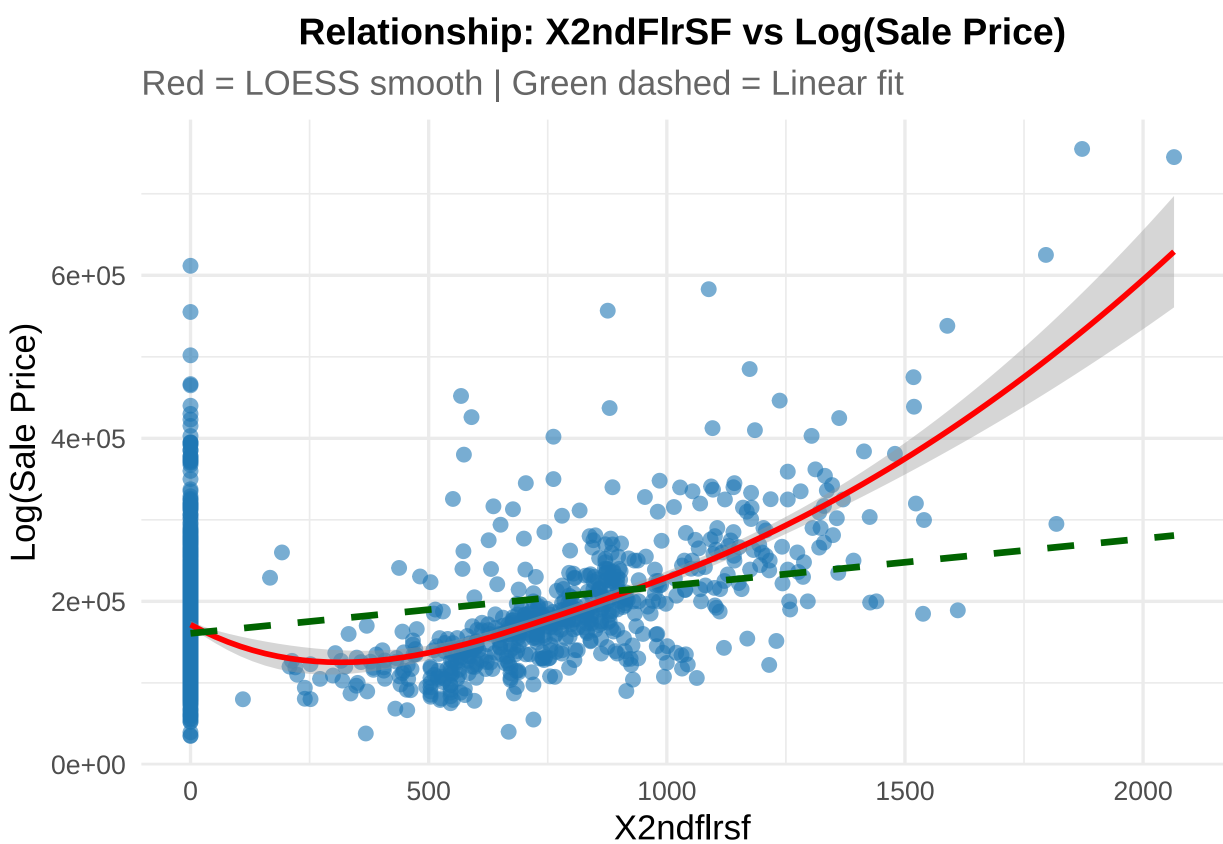
Some evidence of multicollinearity in our model, 5 of the variables with a variance inflation factor (VIF) >= 5, hence we should consider which variables these are and remove the colinear variables.

And finally, a look at the Normality of the residuals shows us that they are fairly normal expect for the tails where there is some clear deviation. Once we address the non-linear relationships, remove the influential observation and the collinear variables, this plot should follow the line more closely.

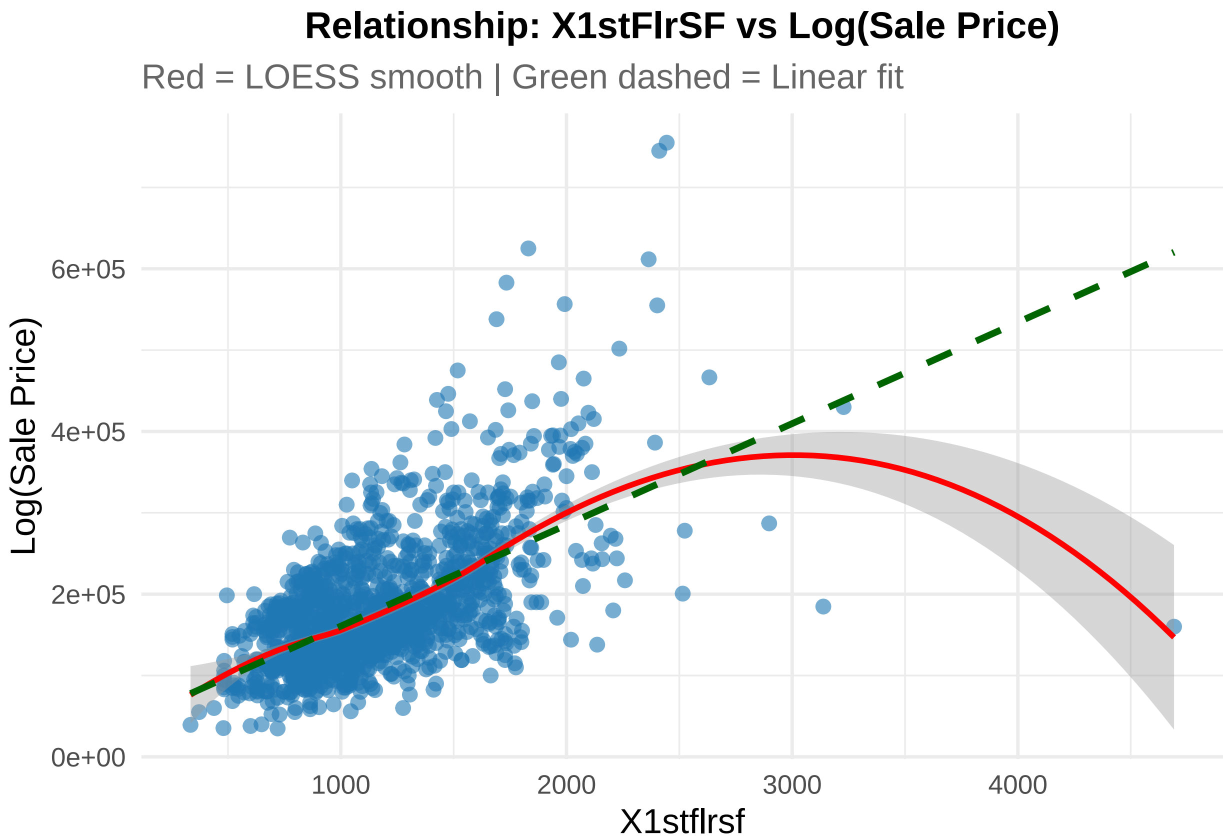
Addressing Model Assumption Findings

As there are a few too many variables to show all of the plot I will show the variables which showes a clear non-linear relationship with the log of sale price.

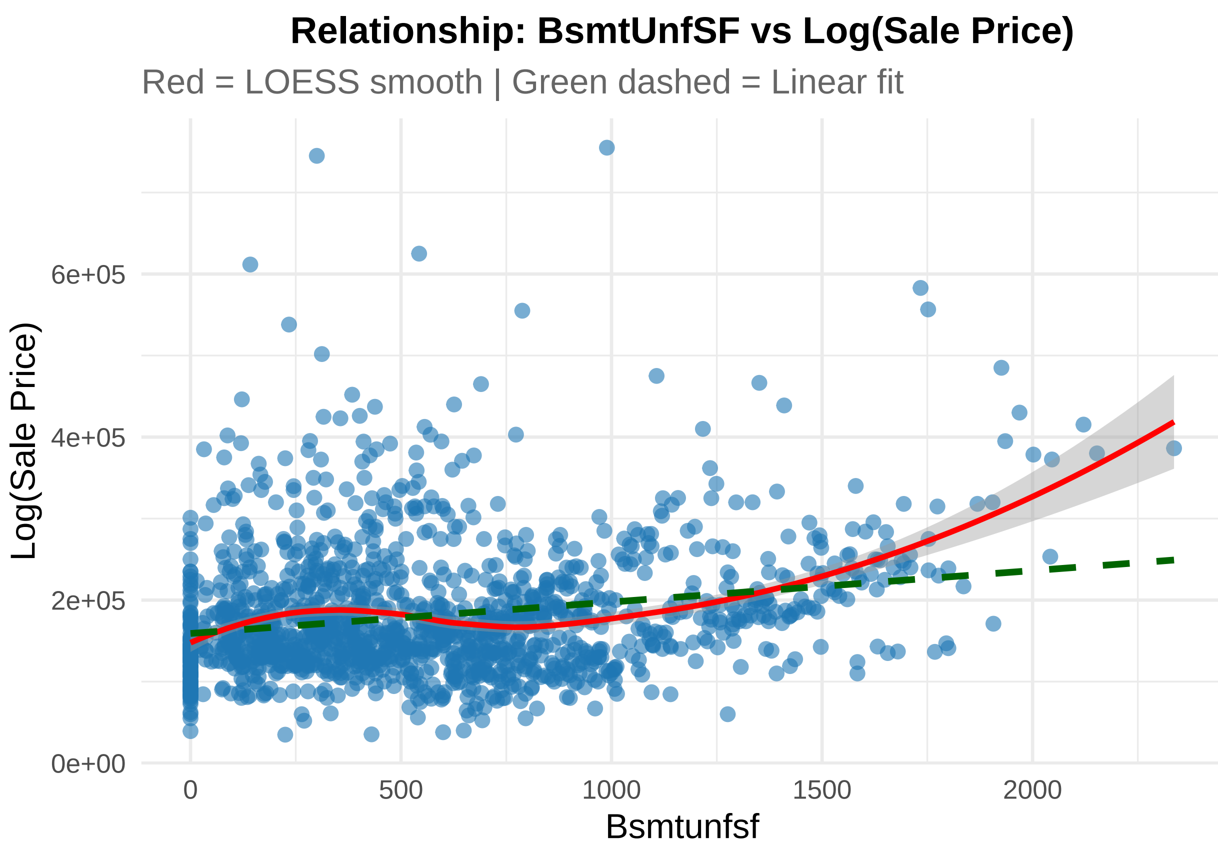
```
## [[1]]
```



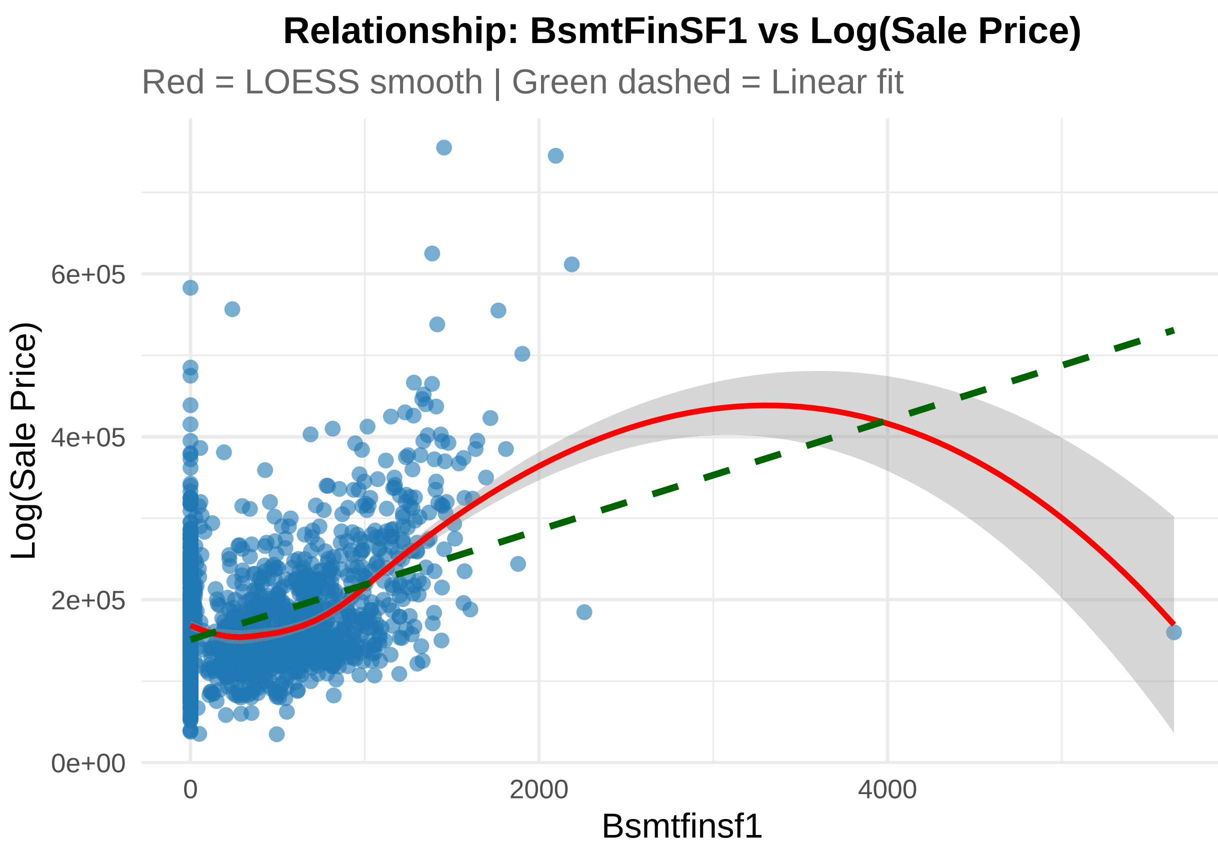

```
##
## [[2]]
```


Relationship: X1stFlrSF vs Log(Sale Price)
Red = LOESS smooth | Green dashed = Linear fit
Log(Sale Price)
6e+05
4e+05
2e+05
0e+00
1000
2000
3000
4000
X1stflrsf

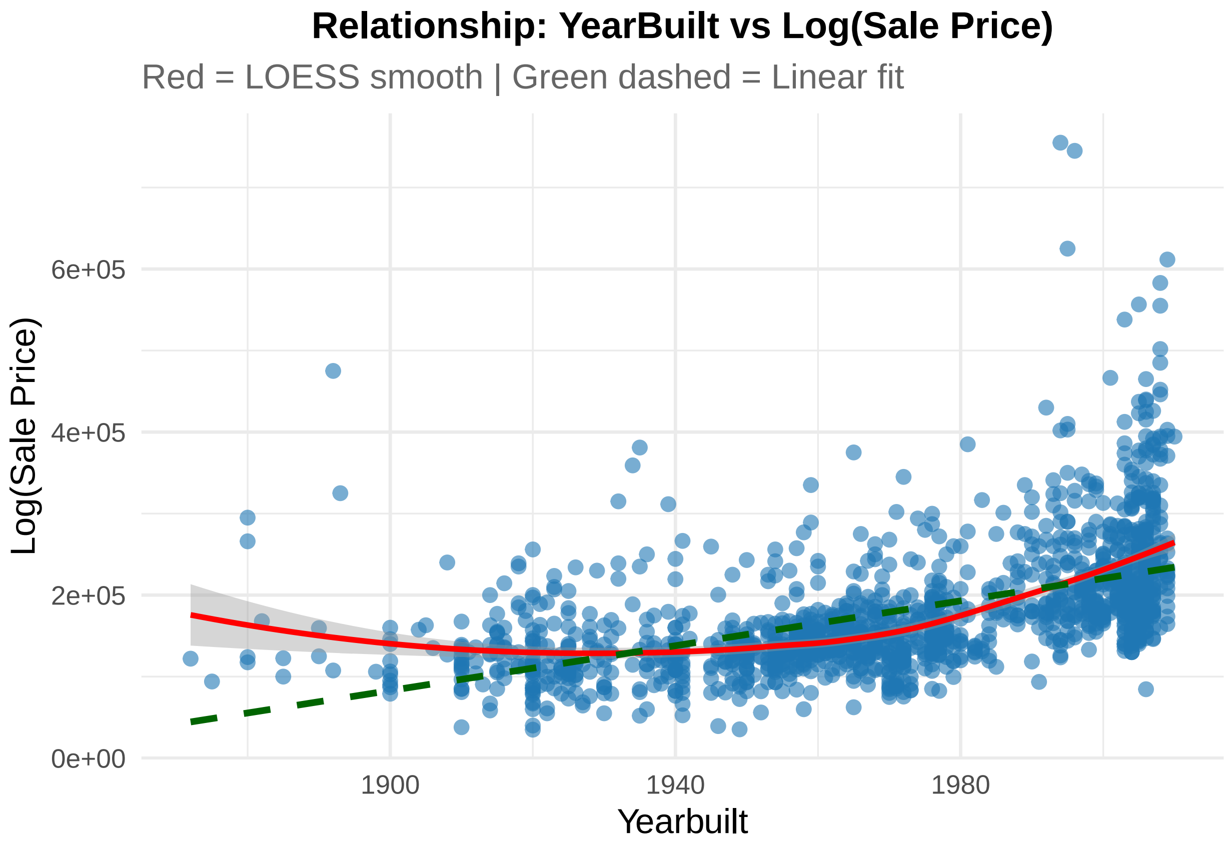
```
##
## [[3]]
```


Relationship: BsmtUnfSF vs Log(Sale Price)
Red = LOESS smooth | Green dashed = Linear fit
Log(Sale Price)
6e+05
4e+05
2e+05
0e+00
0
500
1000
1500
2000
Bsmtunfsf

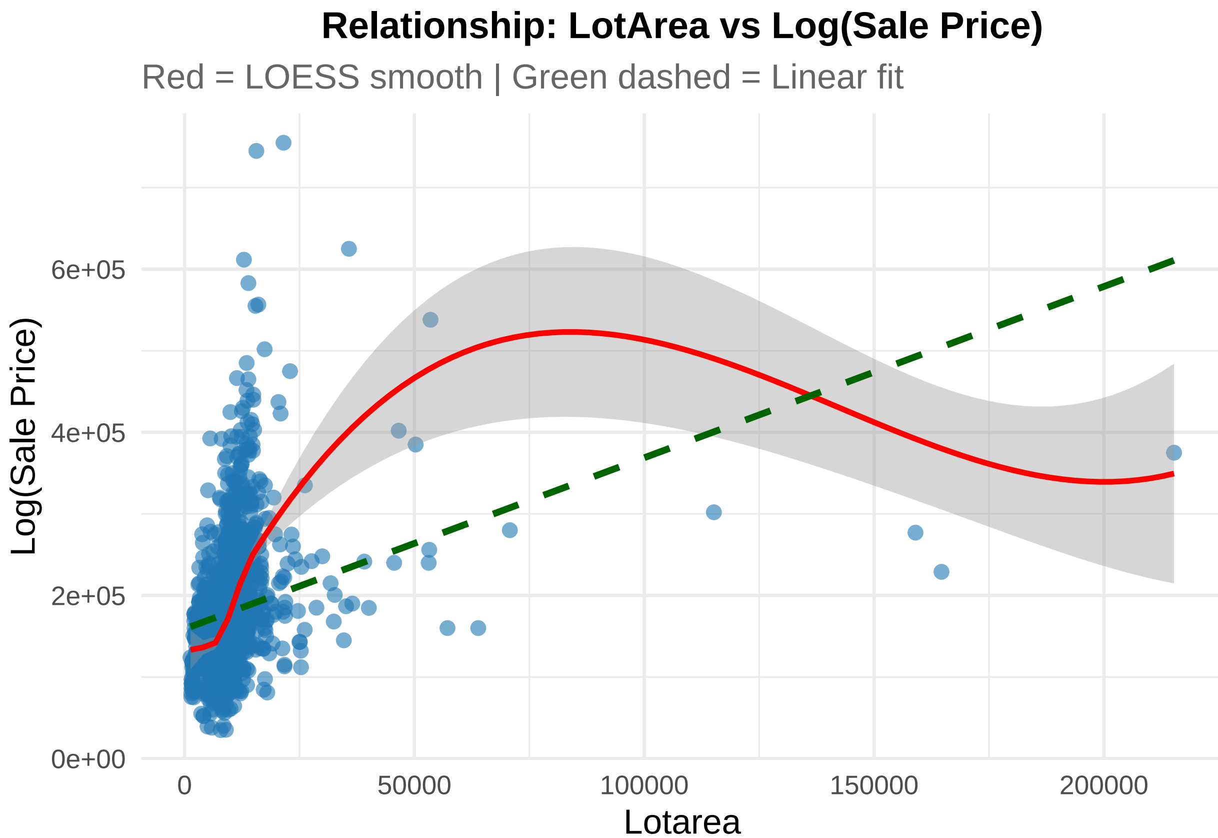
```
##
## [[4]]
```


Relationship: BsmtFinSF1 vs Log(Sale Price)
Red = LOESS smooth | Green dashed = Linear fit
Log(Sale Price)
6e+05
4e+05
2e+05
0e+00
0
2000
4000
Bsmtfinsf1

```
##
## [[5]]
```

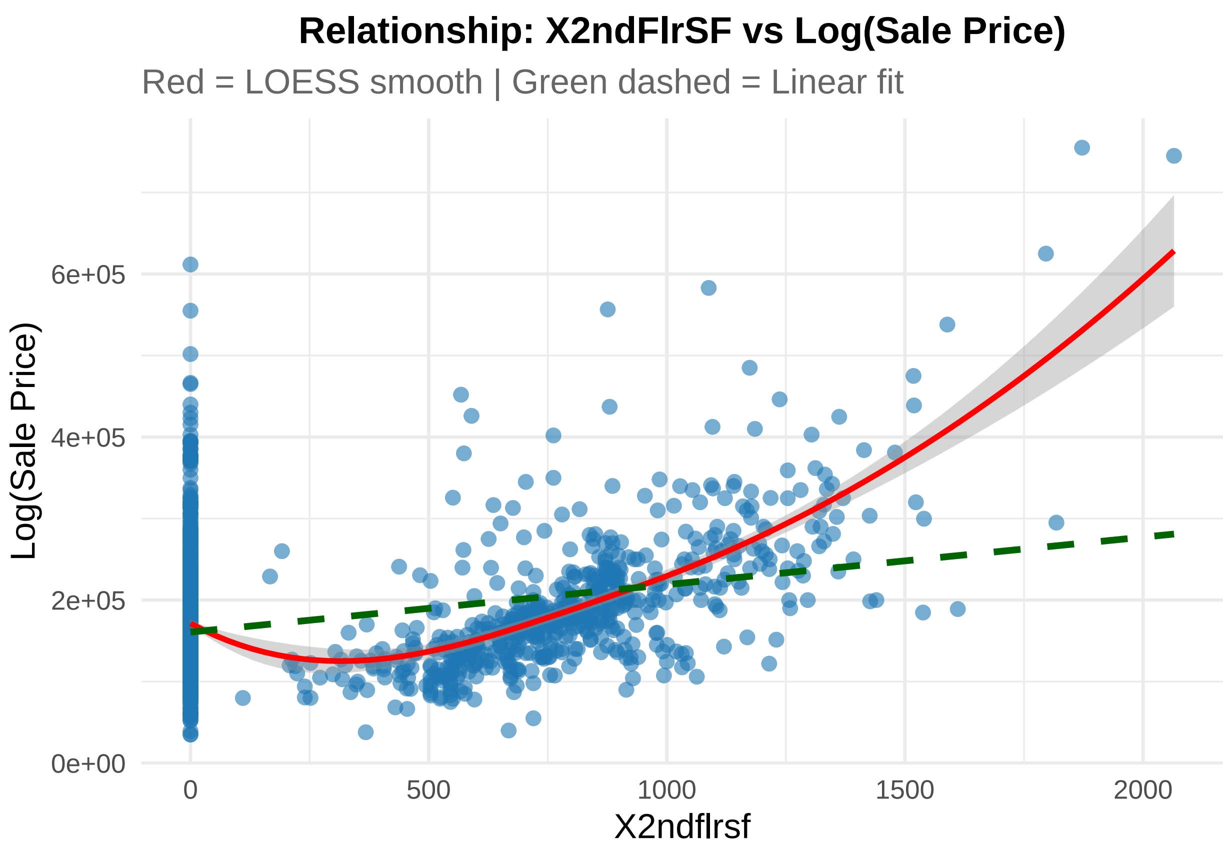

Relationship: YearBuilt vs Log(Sale Price)
Red = LOESS smooth | Green dashed = Linear fit
Log(Sale Price)
6e+05
4e+05
2e+05
0e+00
1900
1940
1980
Yearbuilt

```
##
## [[6]]
```

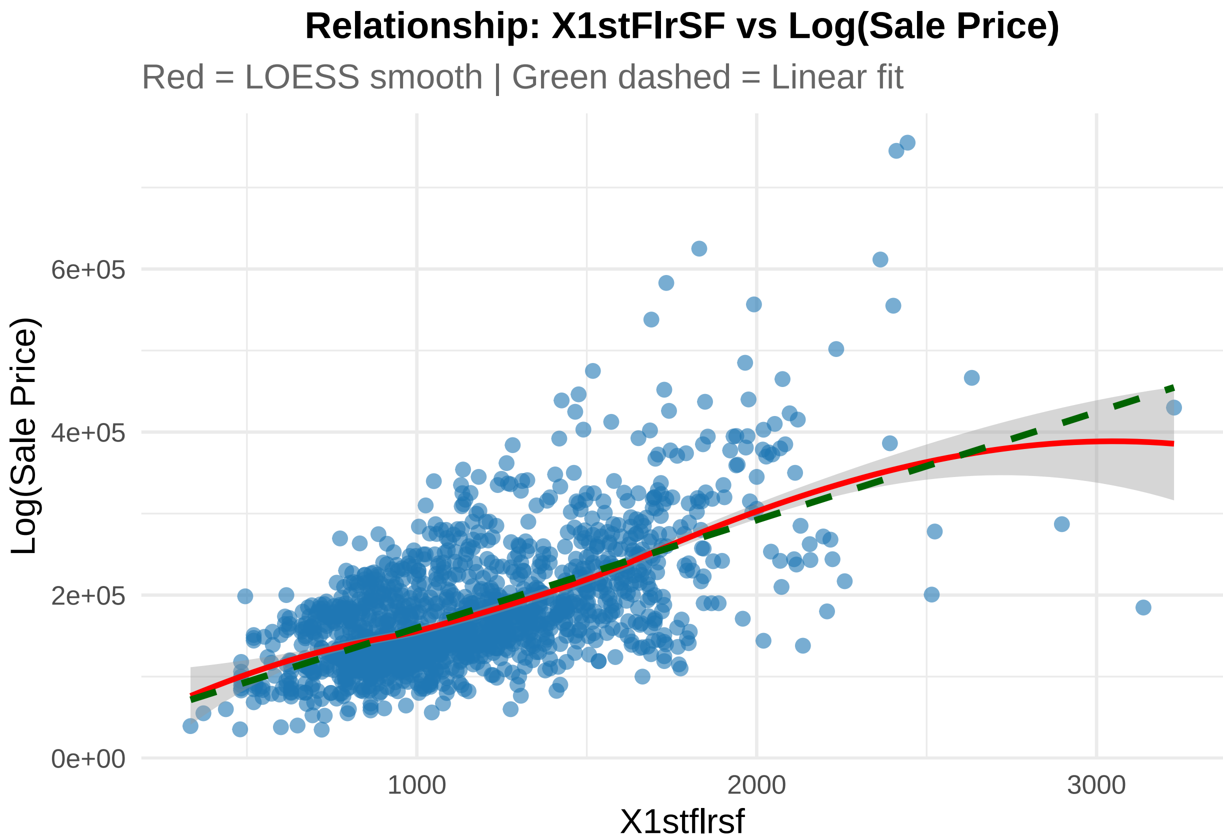


As we can see, there are some clearly non-linear relationships that need to be taken care of to improve our model and meet the required assumptions of a linear model. But a few of the plots show a non-linear relationship due to one observation dragging the line away, hence we should take care to remove this observation and check for linearity again.

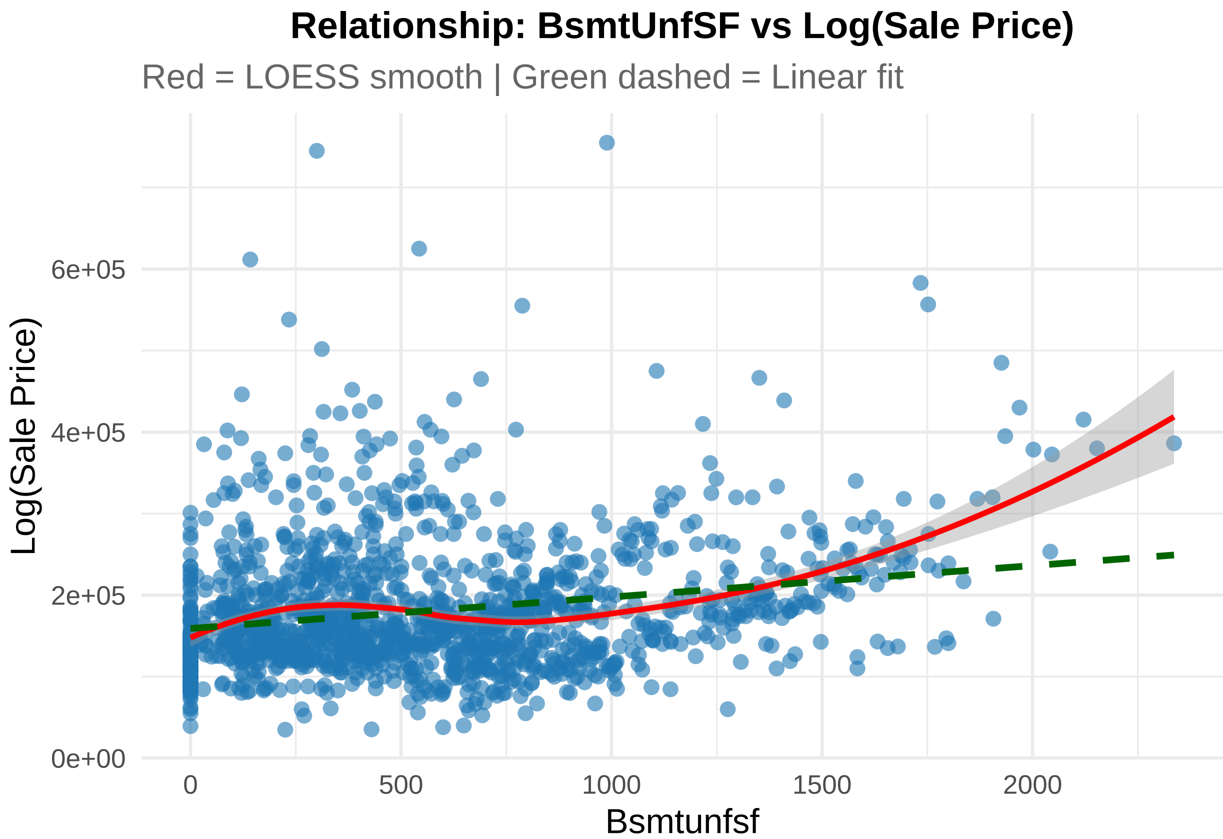
```
## [[1]]
```


Relationship: X2ndFlrSF vs Log(Sale Price)
Red = LOESS smooth | Green dashed = Linear fit
Log(Sale Price)
6e+05
4e+05
2e+05
0e+00
0
500
1000
1500
2000
X2ndflrsf

```
##
## [[2]]
```

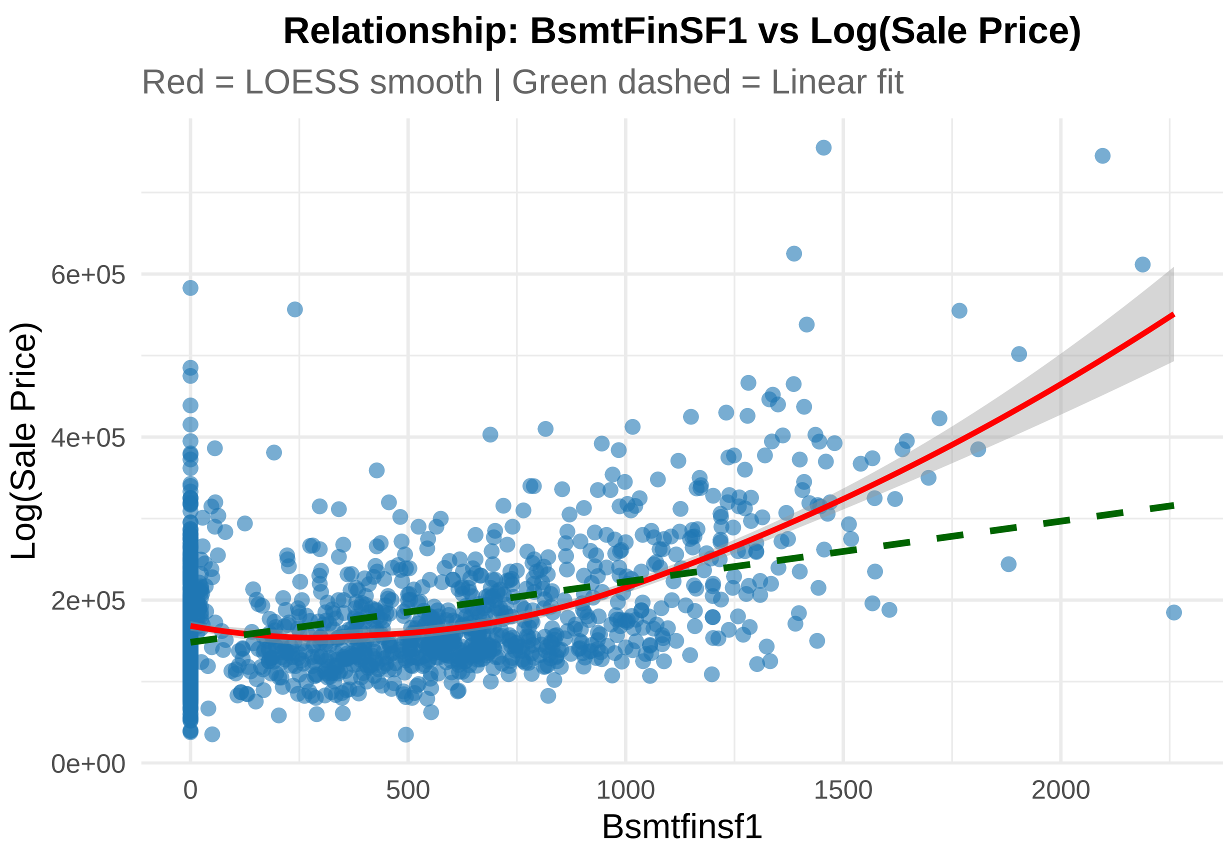

Relationship: X1stFlrSF vs Log(Sale Price)
Red = LOESS smooth | Green dashed = Linear fit
Log(Sale Price)
6e+05
4e+05
2e+05
0e+00
1000
2000
3000
X1stflrsf

```
##
## [[3]]
```

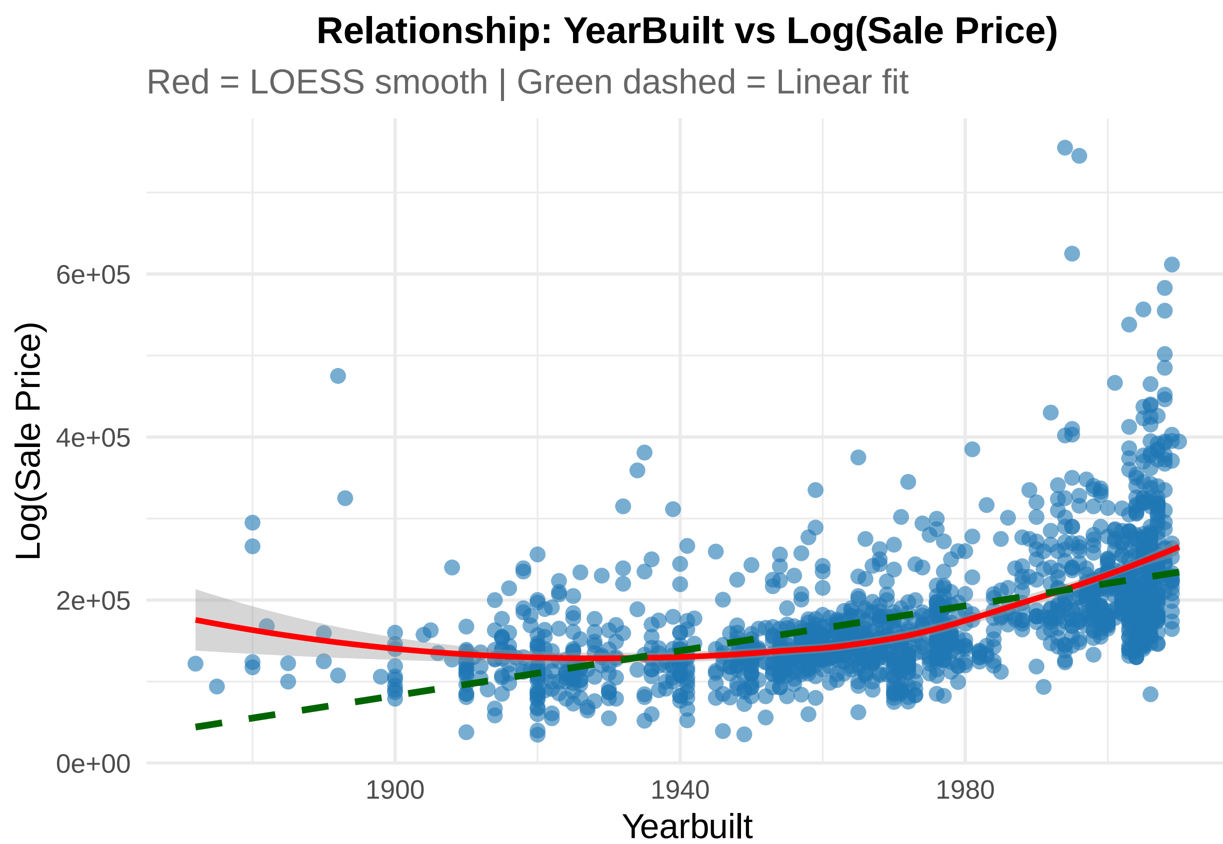


Relationship: BsmtUnfSF vs Log(Sale Price)
Red = LOESS smooth | Green dashed = Linear fit
Log(Sale Price)
6e+05
4e+05
2e+05
0e+00
0
500
1000
1500
2000
Bsmtunfsf

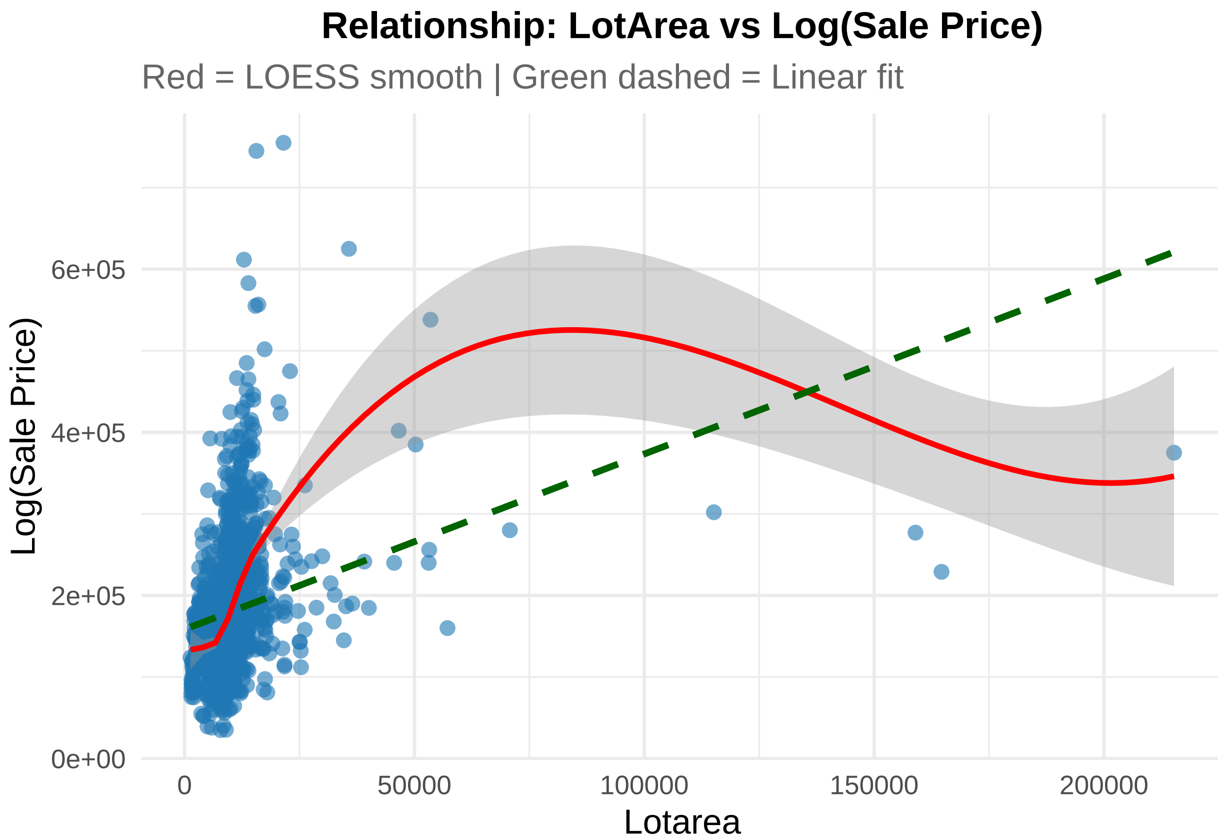

```
##
## [[4]]
```


Relationship: BsmtFinSF1 vs Log(Sale Price)
Red = LOESS smooth | Green dashed = Linear fit
Log(Sale Price)
6e+05
4e+05
2e+05
0e+00
0
500
1000
1500
2000
Bsmtfinsf1

```
##
## [[5]]
```


Relationship: YearBuilt vs Log(Sale Price)
Red = LOESS smooth | Green dashed = Linear fit
Log(Sale Price)
6e+05
4e+05
2e+05
0e+00
1900
1940
1980
Yearbuilt

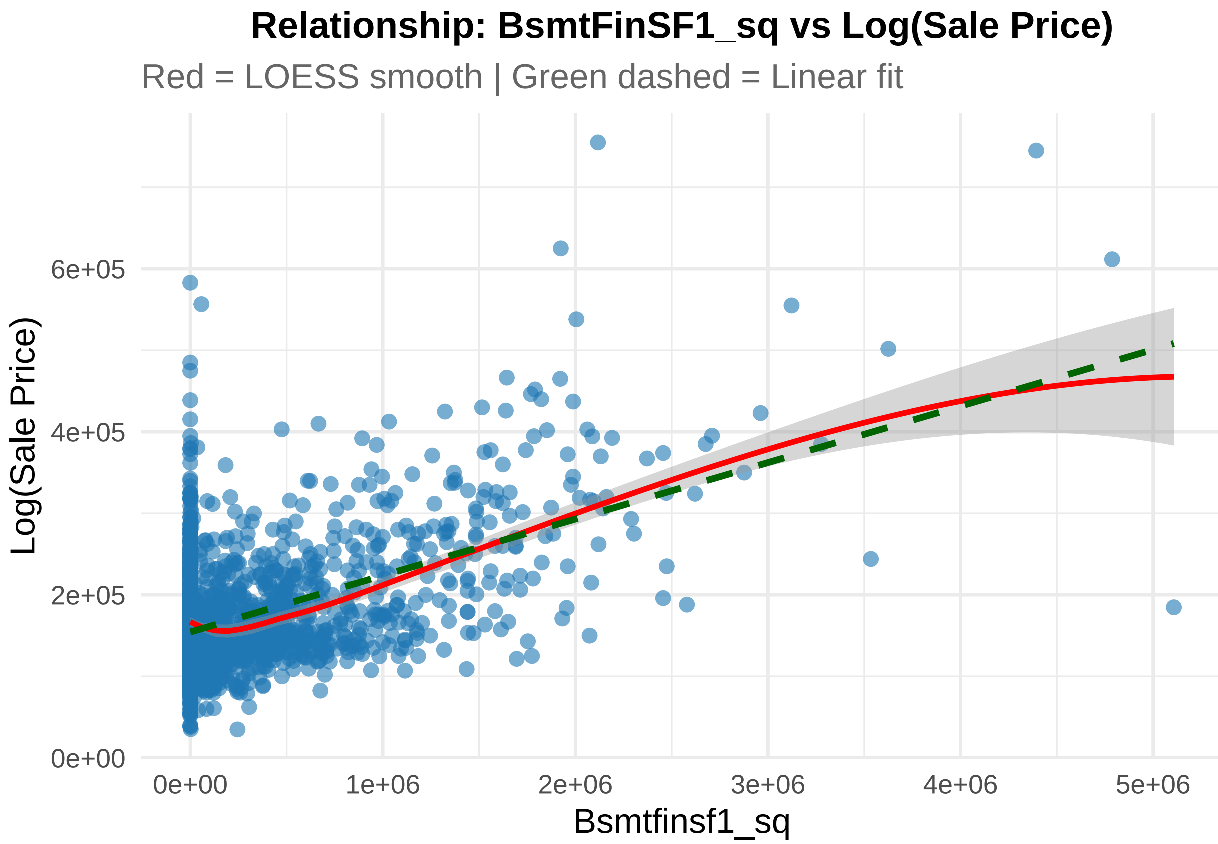
```
##
## [[6]]
```



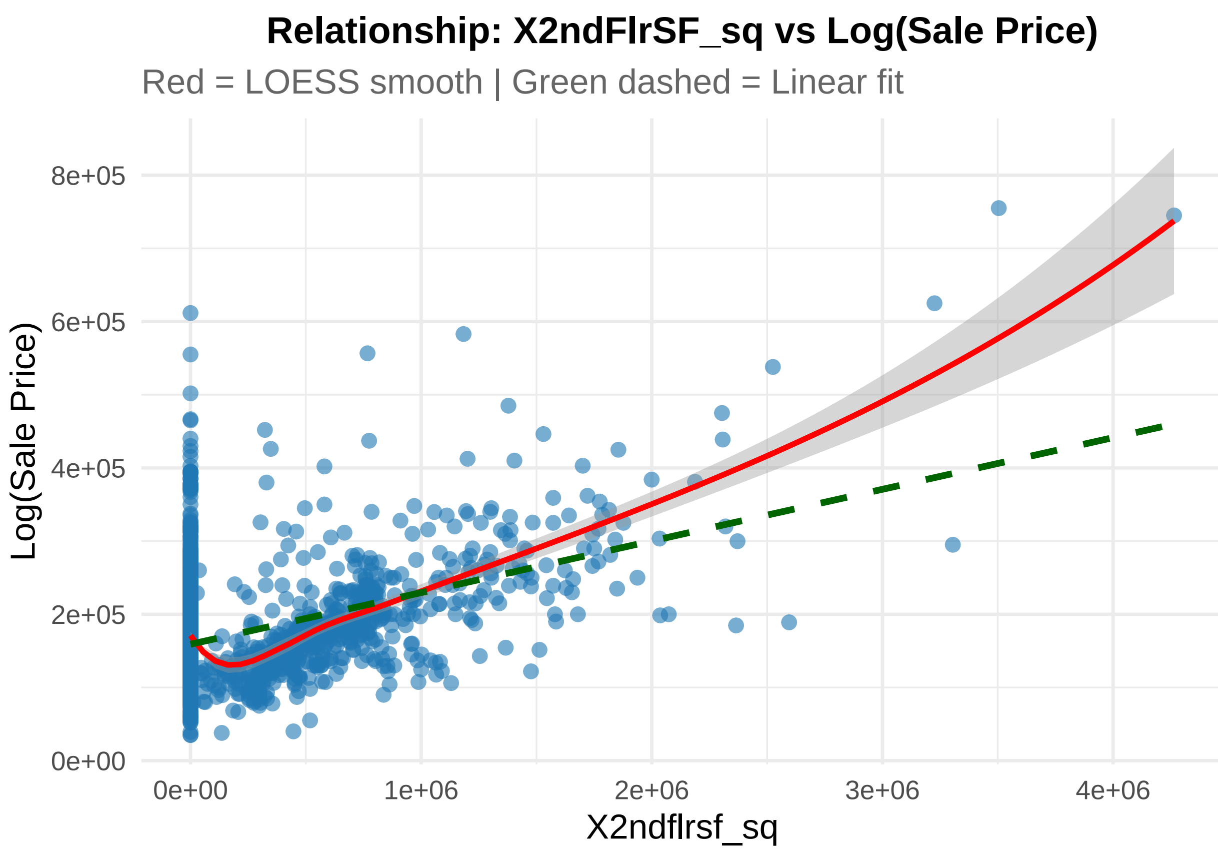
Removing the influential observation, was a wise choice, it was changing the relationship drastically. Regardless, these variables still have a nonlinear relationship which should be taken care of.

LotArea should definitely be logged as some houses have over 100000 square foot lot which influence the relationship greatly and we will put a quadratic on all other variables here except for X1.

```
## [[1]]
```

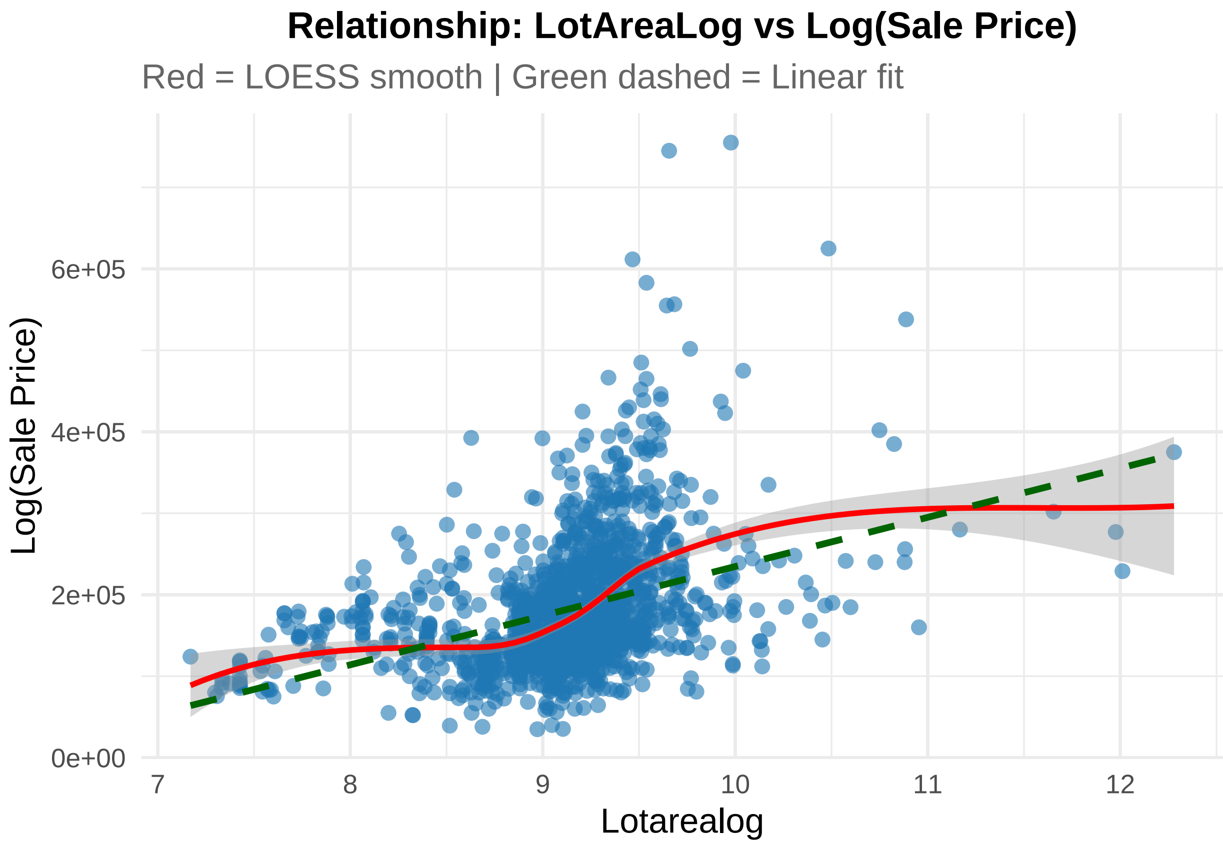

Relationship: BsmtFinSF1_sq vs Log(Sale Price)
Red = LOESS smooth | Green dashed = Linear fit
Log(Sale Price)
6e+05
4e+05
2e+05
0e+00
0e+00
1e+06
2e+06
3e+06
4e+06
5e+06
Bsmtfinsf1_sq

```
##
## [[2]]
```



Relationship: X2ndFlrSF_sq vs Log(Sale Price)
Red = LOESS smooth | Green dashed = Linear fit
Log(Sale Price)
8e+05
6e+05
4e+05
2e+05
0e+00
0e+00
1e+06
2e+06
3e+06
4e+06
X2ndflrsf_sq


```
##
## [[3]]
```



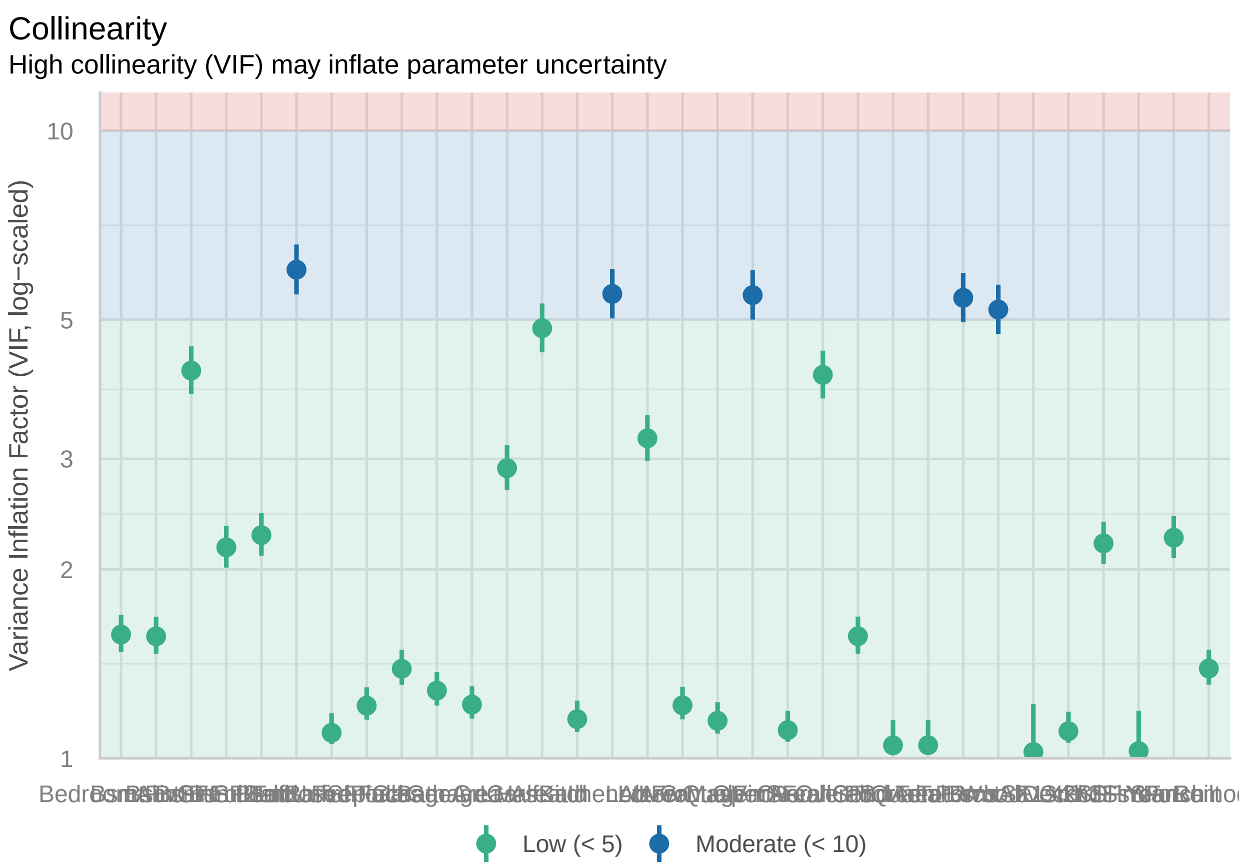
This is definitely an improvement, however it appears some observations are still making the relationships askew, but we will check if they are influecing the model afterwards. Now we will consider multicollinearity.

```
#more details
collinearity <- check_collinearity(model)
print(collinearity)
```

```
## # Check for Multicollinearity
##
## Low Correlation
##
##           Term  VIF   VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
##             Id 1.15 [1.10, 1.24]     1.07      0.87     [0.81, 0.91]
##    LotFrontage 1.21 [1.15, 1.30]     1.10      0.82     [0.77, 0.87]
##        LotArea 3.24 [2.98, 3.53]     1.80      0.31     [0.28, 0.34]
##    OverallQual 1.56 [1.47, 1.68]     1.25      0.64     [0.59, 0.68]
##    OverallCond 4.08 [3.74, 4.46]     2.02      0.24     [0.22, 0.27]
##      YearBuilt 2.25 [2.08, 2.43]     1.50      0.45     [0.41, 0.48]
##   YearRemodAdd 1.39 [1.31, 1.49]     1.18      0.72     [0.67, 0.76]
##     BsmtFinSF1 1.56 [1.47, 1.68]     1.25      0.64     [0.59, 0.68]
##     BsmtFinSF2 4.15 [3.80, 4.53]     2.04      0.24     [0.22, 0.26]
##      X1stFlrSF 1.10 [1.06, 1.19]     1.05      0.91     [0.84, 0.94]
##      X2ndFlrSF 2.20 [2.04, 2.38]     1.48      0.45     [0.42, 0.49]
##   LowQualFinSF 1.15 [1.09, 1.23]     1.07      0.87     [0.81, 0.91]
##      GrLivArea 2.90 [2.67, 3.15]     1.70      0.34     [0.32, 0.37]
##   BsmtFullBath 2.17 [2.01, 2.35]     1.47      0.46     [0.43, 0.50]
##   BsmtHalfBath 2.27 [2.10, 2.46]     1.51      0.44     [0.41, 0.48]
```

```
##        FullBath 1.39 [1.31, 1.49]      1.18      0.72      [0.67, 0.76]
##        HalfBath 4.85 [4.44, 5.31]      2.20      0.21      [0.19, 0.23]
##    BedroomAbvGr 1.57 [1.48, 1.69]      1.25      0.64      [0.59, 0.68]
##      Fireplaces 1.21 [1.15, 1.30]      1.10      0.82      [0.77, 0.87]
##      GarageCars 1.22 [1.16, 1.30]      1.10      0.82      [0.77, 0.86]
##      GarageArea 1.28 [1.21, 1.37]      1.13      0.78      [0.73, 0.82]
##      WoodDeckSF 1.02 [1.00, 1.22]      1.01      0.98      [0.82, 1.00]
##     OpenPorchSF 1.11 [1.06, 1.19]      1.05      0.90      [0.84, 0.94]
##   EnclosedPorch 1.10 [1.05, 1.18]      1.05      0.91      [0.85, 0.95]
##      X3SsnPorch 1.03 [1.00, 1.19]      1.01      0.97      [0.84, 1.00]
##     ScreenPorch 1.05 [1.02, 1.15]      1.02      0.95      [0.87, 0.98]
##        PoolArea 1.05 [1.02, 1.15]      1.02      0.95      [0.87, 0.98]
##
## Moderate Correlation
##
##          Term  VIF   VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
##    MasVnrArea 5.47 [5.00, 6.00]      2.34      0.18      [0.17, 0.20]
##     BsmtUnfSF 6.01 [5.49, 6.59]      2.45      0.17      [0.15, 0.18]
##   TotalBsmtSF 5.42 [4.95, 5.94]      2.33      0.18      [0.17, 0.20]
##  KitchenAbvGr 5.50 [5.02, 6.02]      2.34      0.18      [0.17, 0.20]
##  TotRmsAbvGrd 5.19 [4.75, 5.68]      2.28      0.19      [0.18, 0.21]
```

```r
# Nice plot
plot(collinearity)
```

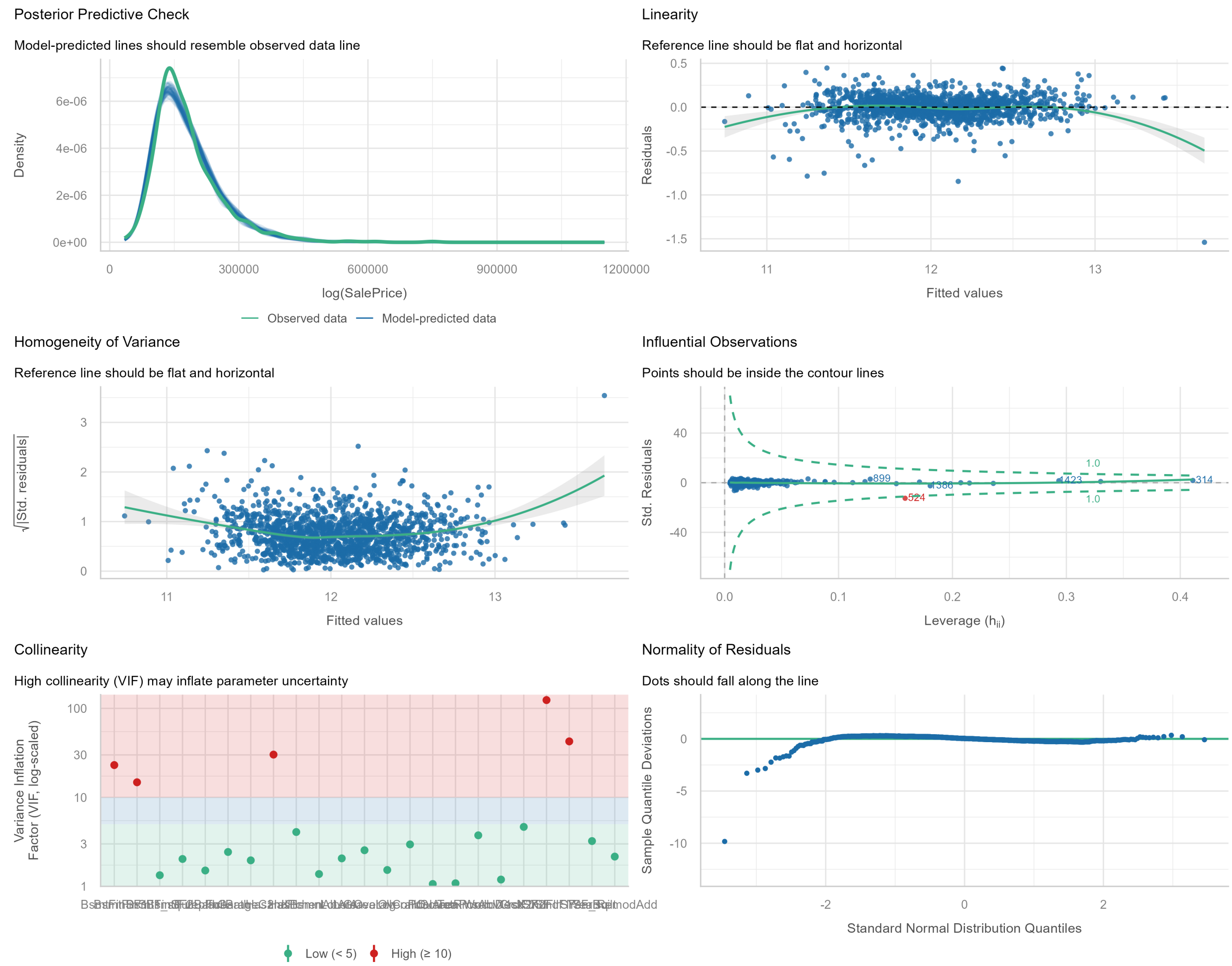


We should now remove one of the terms which have a moderate correlation but is also the least likely to be important to the model. BsmtUnfSF is the total square foot area of unfinished basement, it has the highest correlation and it may be useful but I will make the decision to remove it as this would perhaps only be important for very expensive houses who have unfinished work in the house.

```
modelnew <- lm(log(SalePrice) ~ ., data = num_vars_df[,-c(1,11,12,16,33,34,35,28,18,26,20,30,29,15,2,8,2
summary(modelnew)
```

```
##
## Call:
## lm(formula = log(SalePrice) ~ ., data = num_vars_df[, -c(1, 11,
##     12, 16, 33, 34, 35, 28, 18, 26, 20, 30, 29, 15, 2, 8, 21)])
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.54005 -0.06190  0.00491  0.07221  0.44678
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.449e+00  5.002e-01   2.898 0.003816 **
## LotArea       -2.811e-07  5.094e-07  -0.552 0.581187
## OverallQual    8.751e-02  4.372e-03  20.015  < 2e-16 ***
## OverallCond    4.534e-02  3.893e-03  11.645  < 2e-16 ***
## YearBuilt      2.889e-03  2.086e-04  13.848  < 2e-16 ***
## YearRemodAdd   1.345e-03  2.496e-04   5.389 8.28e-08 ***
## BsmtFinSF1     1.019e-04  3.866e-05   2.635 0.008502 **
## BsmtFinSF2     2.193e-05  2.510e-05   0.874 0.382329
## X1stFlrSF      2.964e-04  2.015e-05  14.707  < 2e-16 ***
## X2ndFlrSF      3.428e-04  8.941e-05   3.834 0.000132 ***
## BsmtFullBath   2.898e-02  9.658e-03   3.001 0.002741 **
## FullBath       1.239e-02  9.928e-03   1.248 0.212237
## KitchenAbvGr  -9.166e-02  1.866e-02  -4.912 1.00e-06 ***
## TotRmsAbvGrd   7.957e-03  4.190e-03   1.899 0.057766 .
## Fireplaces     3.428e-02  6.708e-03   5.111 3.63e-07 ***
## GarageCars     5.788e-02  6.567e-03   8.814  < 2e-16 ***
## WoodDeckSF     6.603e-05  3.055e-05   2.161 0.030824 *
## ScreenPorch    2.474e-04  6.542e-05   3.782 0.000162 ***
## PoolArea       4.516e-05  9.466e-05   0.477 0.633382
## LotAreaLog     9.281e-02  1.086e-02   8.544  < 2e-16 ***
## Has2ndFlr     -5.536e-02  3.890e-02  -1.423 0.154944
## X2ndFlrSF_sq  -7.038e-08  4.689e-08  -1.501 0.133605
## HasBsmnt       8.406e-03  1.518e-02   0.554 0.579698
## BsmtFinSF1_sq -2.232e-08  2.330e-08  -0.958 0.338299
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1339 on 1435 degrees of freedom
## Multiple R-squared:  0.8896, Adjusted R-squared:  0.8878
## F-statistic: 502.5 on 23 and 1435 DF,  p-value: < 2.2e-16
```

```
## Ignoring unknown labels:
## * size : ""
```



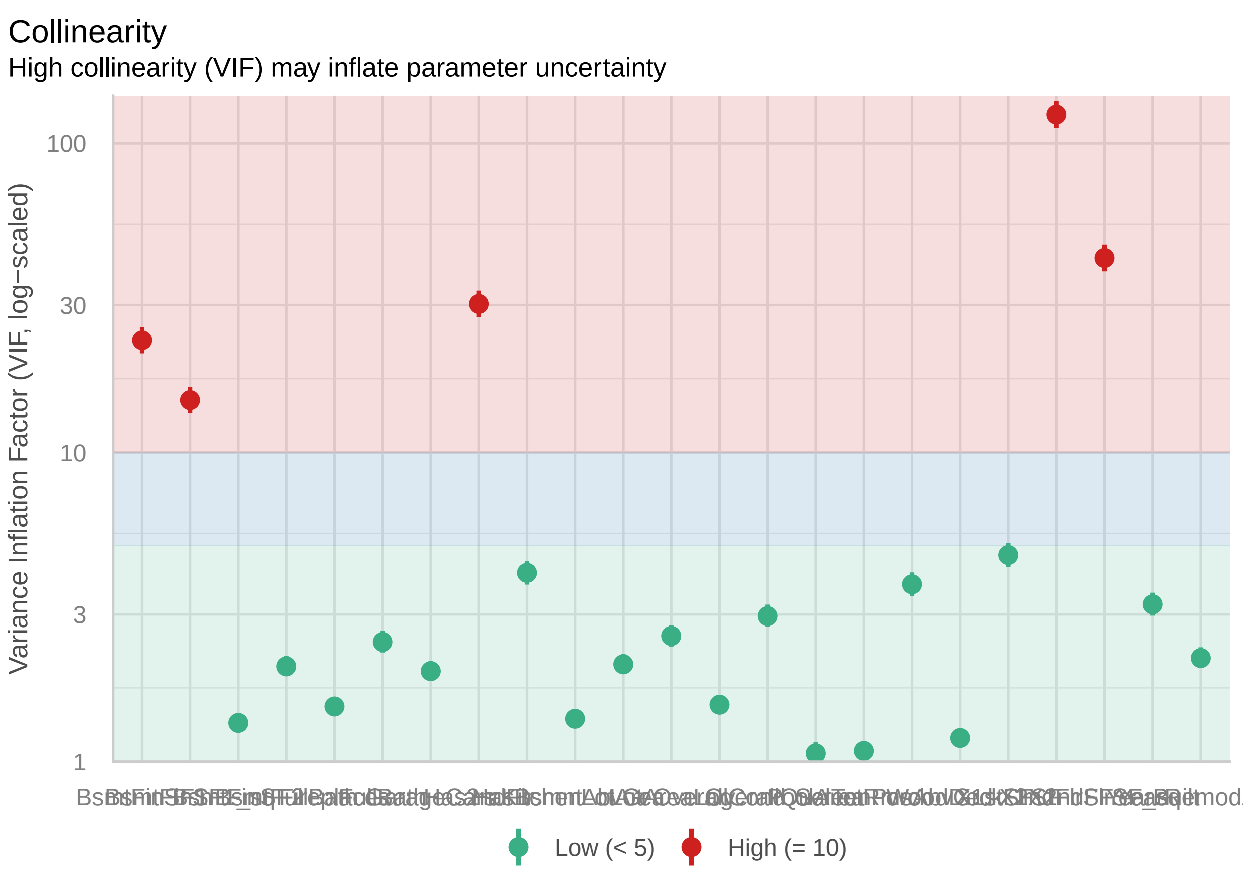
Great improvement in the linear model assumptions, we can see that the model prediction is very close to the observed data across all ranges now but we have introduced some strong multicollinearity but this is due to the non-linear predictors we added to the model, but to confirm this, lets check it. We should also remove observation 524, as it clearly is outside of the range of expected values.

```
collinearity <- check_collinearity(modelnew)
print(collinearity)
```

```
## # Check for Multicollinearity
##
## Low Correlation
##
##           Term  VIF      VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
##        LotArea 2.06 [  1.92,    2.23]     1.44      0.48     [0.45, 0.52]
##    OverallQual 2.96 [  2.73,    3.23]     1.72      0.34     [0.31, 0.37]
##    OverallCond 1.53 [  1.43,    1.64]     1.24      0.65     [0.61, 0.70]
##      YearBuilt 3.23 [  2.97,    3.52]     1.80      0.31     [0.28, 0.34]
##   YearRemodAdd 2.16 [  2.01,    2.34]     1.47      0.46     [0.43, 0.50]
##     BsmtFinSF2 1.33 [  1.26,    1.43]     1.16      0.75     [0.70, 0.79]
##      X1stFlrSF 4.66 [  4.26,    5.10]     2.16      0.21     [0.20, 0.23]
##   BsmtFullBath 2.03 [  1.89,    2.20]     1.43      0.49     [0.45, 0.53]
##       FullBath 2.44 [  2.25,    2.64]     1.56      0.41     [0.38, 0.44]
```

```
##   KitchenAbvGr 1.38 [   1.30,    1.48]      1.17       0.73      [0.68, 0.77]
##   TotRmsAbvGrd 3.75 [   3.44,    4.10]      1.94       0.27      [0.24, 0.29]
##     Fireplaces 1.51 [   1.42,    1.62]      1.23       0.66      [0.62, 0.71]
##     GarageCars 1.96 [   1.82,    2.12]      1.40       0.51      [0.47, 0.55]
##     WoodDeckSF 1.19 [   1.13,    1.28]      1.09       0.84      [0.78, 0.88]
##    ScreenPorch 1.08 [   1.04,    1.17]      1.04       0.92      [0.86, 0.96]
##       PoolArea 1.06 [   1.03,    1.16]      1.03       0.94      [0.87, 0.97]
##     LotAreaLog 2.55 [   2.35,    2.77]      1.60       0.39      [0.36, 0.42]
##       HasBsmnt 4.08 [   3.74,    4.46]      2.02       0.25      [0.22, 0.27]
##
## High Correlation
##
##           Term    VIF       VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
##     BsmtFinSF1  23.07 [ 20.90,  25.46]     4.80      0.04      [0.04, 0.05]
##      X2ndFlrSF 123.89 [112.06, 136.97]    11.13  8.07e-03      [0.01, 0.01]
##      Has2ndFlr  30.24 [ 27.39,  33.40]     5.50      0.03      [0.03, 0.04]
##    X2ndFlrSF_sq  42.56 [ 38.53,  47.02]     6.52      0.02      [0.02, 0.03]
##   BsmtFinSF1_sq  14.77 [ 13.41,  16.29]     3.84      0.07      [0.06, 0.07]
```

```
# Nice plot
plot(collinearity)
```



As expected, the high correlation variables are the non-linear terms and the logical variables, we will not remove them for that reason.

```r
#Removing the influential observation
num_vars_df <- num_vars_df[-524,]

modelnew <- lm(log(SalePrice) ~ ., data = num_vars_df[,-c(1,11,12,16,33,34,35,28,18,26,20,30,29,15,2,8,2
summary(modelnew)
```

```
##
## Call:
## lm(formula = log(SalePrice) ~ ., data = num_vars_df[, -c(1, 11,
##     12, 16, 33, 34, 35, 28, 18, 26, 20, 30, 29, 15, 2, 8, 21)])
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.82756 -0.05821  0.00474  0.07160  0.46212
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.442e+00  4.721e-01   3.054 0.002300 **
## LotArea       -1.076e-07  4.810e-07  -0.224 0.823023
## OverallQual    8.410e-02  4.135e-03  20.338  < 2e-16 ***
## OverallCond    4.766e-02  3.679e-03  12.954  < 2e-16 ***
## YearBuilt      3.018e-03  1.972e-04  15.305  < 2e-16 ***
## YearRemodAdd   1.229e-03  2.358e-04   5.210 2.16e-07 ***
## BsmtFinSF1    -2.299e-05  3.769e-05  -0.610 0.541888
## BsmtFinSF2     1.128e-05  2.370e-05   0.476 0.634389
## X1stFlrSF      3.239e-04  1.913e-05  16.928  < 2e-16 ***
## X2ndFlrSF      3.831e-04  8.445e-05   4.536 6.20e-06 ***
## BsmtFullBath   2.877e-02  9.116e-03   3.155 0.001636 **
## FullBath       9.850e-03  9.374e-03   1.051 0.293497
## KitchenAbvGr  -9.907e-02  1.762e-02  -5.622 2.27e-08 ***
## TotRmsAbvGrd   3.726e-03  3.968e-03   0.939 0.347894
## Fireplaces     2.731e-02  6.353e-03   4.298 1.84e-05 ***
## GarageCars     5.412e-02  6.206e-03   8.722  < 2e-16 ***
## WoodDeckSF     5.209e-05  2.885e-05   1.805 0.071257 .
## ScreenPorch    2.333e-04  6.176e-05   3.778 0.000165 ***
## PoolArea      -1.186e-05  8.946e-05  -0.133 0.894578
## LotAreaLog     9.299e-02  1.025e-02   9.069  < 2e-16 ***
## Has2ndFlr     -6.468e-02  3.673e-02  -1.761 0.078450 .
## X2ndFlrSF_sq  -6.878e-08  4.426e-08  -1.554 0.120459
## HasBsmnt       3.785e-02  1.449e-02   2.611 0.009123 **
## BsmtFinSF1_sq  7.164e-08  2.310e-08   3.101 0.001966 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1263 on 1434 degrees of freedom
## Multiple R-squared:  0.9017, Adjusted R-squared:  0.9001
## F-statistic: 571.7 on 23 and 1434 DF,  p-value: < 2.2e-16
```

Our model is getting better and better. Now we can explain 90.17% of the variation in logged Sale Prices however we have two insignificant variables which we will remove only BsmtFinSF2 as the other is the a member of the non-linear variables we created before.

```r
modelnew2 <- lm(log(SalePrice) ~ ., data = num_vars_df[,-c(1,10,11,12,16,33,34,35,28,18,26,20,30,29,15,2
summary(modelnew2)
```

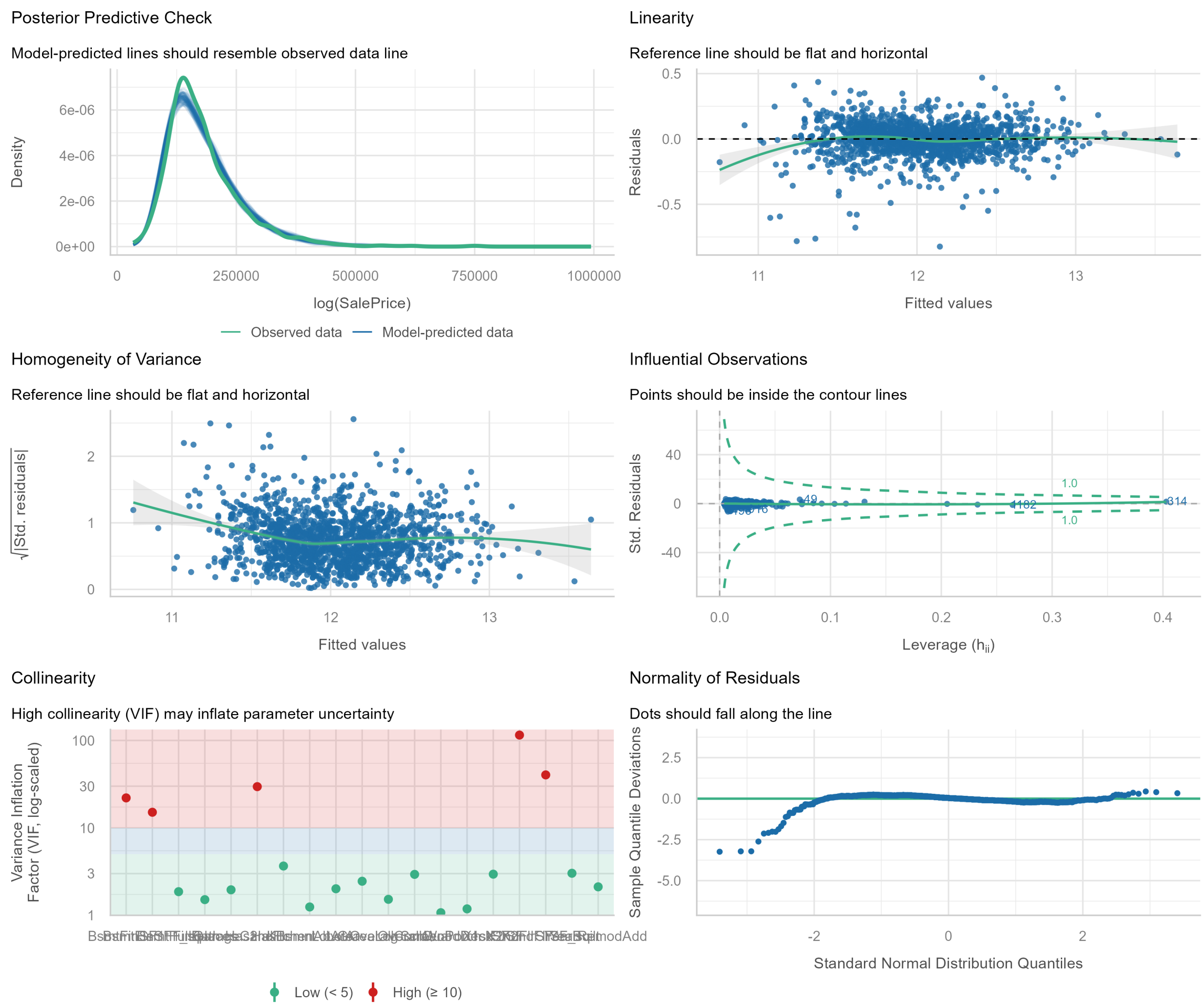
```
##
## Call:
## lm(formula = log(SalePrice) ~ ., data = num_vars_df[, -c(1, 10,
##     11, 12, 16, 33, 34, 35, 28, 18, 26, 20, 30, 29, 15, 2, 8,
##     21)])
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.82714 -0.05864  0.00484  0.07094  0.46214
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.452e+00  4.715e-01   3.079 0.002119 **
## LotArea       -9.253e-08  4.798e-07  -0.193 0.847113
## OverallQual    8.400e-02  4.129e-03  20.344  < 2e-16 ***
## OverallCond    4.774e-02  3.675e-03  12.991  < 2e-16 ***
## YearBuilt      3.019e-03  1.971e-04  15.315  < 2e-16 ***
## YearRemodAdd   1.222e-03  2.354e-04   5.193 2.37e-07 ***
## BsmtFinSF1    -2.837e-05  3.594e-05  -0.789 0.430003
## X1stFlrSF      3.251e-04  1.898e-05  17.129  < 2e-16 ***
## X2ndFlrSF      3.858e-04  8.424e-05   4.579 5.07e-06 ***
## BsmtFullBath   2.984e-02  8.832e-03   3.378 0.000749 ***
## FullBath       9.796e-03  9.370e-03   1.045 0.295988
## KitchenAbvGr  -9.934e-02  1.761e-02  -5.642 2.03e-08 ***
## TotRmsAbvGrd   3.641e-03  3.963e-03   0.919 0.358403
## Fireplaces     2.732e-02  6.352e-03   4.301 1.82e-05 ***
## GarageCars     5.407e-02  6.203e-03   8.717  < 2e-16 ***
## WoodDeckSF     5.310e-05  2.877e-05   1.846 0.065152 .
## ScreenPorch    2.357e-04  6.154e-05   3.831 0.000133 ***
## PoolArea      -1.024e-05  8.937e-05  -0.115 0.908781
## LotAreaLog     9.304e-02  1.025e-02   9.078  < 2e-16 ***
## Has2ndFlr     -6.583e-02  3.664e-02  -1.797 0.072616 .
## X2ndFlrSF_sq  -7.000e-08  4.418e-08  -1.585 0.113266
## HasBsmnt       4.024e-02  1.359e-02   2.961 0.003117 **
## BsmtFinSF1_sq  7.338e-08  2.280e-08   3.218 0.001318 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1263 on 1435 degrees of freedom
## Multiple R-squared:  0.9016, Adjusted R-squared:  0.9001
## F-statistic:   598 on 22 and 1435 DF,  p-value: < 2.2e-16
```

The Pool Area is now insignificant, so we will remove variables one by one until all are significant.

Now we have all significant variables in our model to the 10% level at least.

Lets check our assumptions:

```
## Ignoring unknown labels:
## * size : ""
```



Model Diagnostics Summary

Our final model satisfies the core assumptions of linear regression to a high standard:

- **Linearity**: The residuals versus fitted values plot shows a nearly random scatter around zero, indicating that the functional form (including polynomial terms where used) appropriately captures the relationships in the data.
- **Homoscedasticity**: The spread of residuals is largely consistent across fitted values, supporting the assumption of constant variance.
- **Normality of Residuals**: The residuals are approximately normally distributed around zero, with only a minor left skew. This small deviation is unlikely to meaningfully impact model predictions.
- **Multicollinearity**: After centering the polynomial terms, variance inflation factors (VIF) are within acceptable ranges.

Overall, the diagnostic checks confirm that the model is reliable and well-specified. With an Adjusted R^2 of **90.16%**, the model explains a very high proportion of the variation in logged house sale prices.

We are now ready to generate predictions on the test dataset and submit our results to Kaggle.

From the logarithmic model of sales prices using only the numerical variables in the dataset we ended up with a RMSE of 0.15623, placing us in the top 60% only, hence we need to improve our model is not up to scratch.